

## MYS BYKOV, Russian Federation

WMO: 218210

Lat: 72.004N

Lon: 129.118E

Elev: 54

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 84-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 1             | -40.3      | -37.6 | -45.8                         | 0.4 | -39.5 | -43.2 | 0.5 | -37.3 | 38.4                  | -10.0 | 35.1 | -10.9 | 4.2                   | 140  | 1.076 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 7.2                    | 63.1            | 57.2 | 59.2 | 54.6 | 55.9 | 52.2 | 57.6                | 62.2 | 54.9 | 58.6 | 52.4 | 55.9 | 9.7                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 55.0                            | 64.6 | 60.3 | 52.9 | 59.6 | 57.8 | 49.9 | 53.4 | 55.3 | 25.0          | 62.7 | 23.3 | 58.9 | 21.8 | 56.0 | 65.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 32.9              | 28.8 | 26.4 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |      | -43.1                      | 60.6 | 4.7                | 3.2 | -46.5                                              | 62.9 | -49.3      | 64.8 | -51.9      | 66.6 | -55.4      | 68.9 |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov  | Dec  |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)  | (p)  |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 11.8  | -22.4 | -21.1 | -10.1 | 5.1   | 19.2  | 38.6  | 47.0  | 47.5  | 37.6  | 17.2 | -3.2 | -15.5 | (14) |
|      |                                                                   | DBStd       | 27.01 | 8.59  | 10.91 | 14.88 | 11.97 | 8.21  | 7.78  | 5.97  | 5.64  | 6.14  | 9.68 | 9.03 | 9.33  |      |
|      |                                                                   | HDD50       | 14023 | 2244  | 1990  | 1865  | 1347  | 956   | 355   | 133   | 118   | 374   | 1016 | 1596 | 2029  |      |
|      |                                                                   | HDD65       | 19407 | 2709  | 2410  | 2330  | 1797  | 1421  | 793   | 560   | 543   | 823   | 1481 | 2046 | 2494  |      |
|      |                                                                   | CDD50       | 92    | 0     | 0     | 0     | 0     | 0     | 12    | 39    | 40    | 1     | 0    | 0    | 0     |      |
|      |                                                                   | CDD65       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0     |      |
|      |                                                                   | CDH74       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0     |      |
|      |                                                                   | CDH80       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0     |      |
|      |                                                                   | WSAvg       | 11.3  | 10.0  | 10.2  | 10.5  | 9.5   | 10.5  | 12.6  | 12.9  | 12.5  | 12.0  | 12.4 | 12.1 | 10.6  |      |
|      |                                                                   | PrecAvg     | 15.6  | 1.1   | 0.7   | 0.8   | 0.6   | 1.1   | 1.5   | 2.1   | 2.2   | 1.7   | 1.5  | 1.1  | 1.3   |      |
| (15) | Precipitation                                                     | PrecMax     | 22.4  | 2.9   | 2.6   | 2.0   | 2.8   | 2.8   | 4.8   | 5.0   | 4.1   | 2.6   | 1.9  | 3.8  |       | (18) |
| (16) |                                                                   | PrecMin     | 10.7  | 0.2   | 0.0   | 0.2   | 0.0   | 0.2   | 0.4   | 0.7   | 1.1   | 0.6   | 0.5  | 0.4  | 0.1   |      |
| (17) |                                                                   | PrecStd     | 2.8   | 0.7   | 0.6   | 0.5   | 0.6   | 0.6   | 1.0   | 0.9   | 0.8   | 0.6   | 0.4  | 0.9  |       |      |
| (18) |                                                                   |             |       |       |       |       |       |       |       |       |       |       |      |      |       |      |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 1.5   | 9.4   | 26.8  | 32.9  | 40.4  | 66.1  | 68.7  | 67.1  | 54.6  | 35.4  | 22.2 | 8.1  |       | (26) |
| (20) |                                                                   | 0.4% MCWB   | 0.7   | 8.4   | 24.9  | 30.4  | 36.3  | 58.2  | 60.6  | 58.0  | 50.2  | 33.9  | 21.6 | 7.0  |       |      |
| (21) |                                                                   | 2% DB       | -3.3  | 3.4   | 22.4  | 29.0  | 35.8  | 58.7  | 64.3  | 61.9  | 50.9  | 32.7  | 17.0 | 4.4  |       |      |
| (22) |                                                                   | 2% MCWB     | -3.8  | 2.8   | 21.2  | 26.9  | 33.6  | 52.8  | 58.4  | 56.4  | 47.4  | 31.3  | 16.1 | 3.6  |       |      |
| (23) |                                                                   | 5% DB       | -7.0  | -2.6  | 18.1  | 25.6  | 33.9  | 53.8  | 60.1  | 58.5  | 48.6  | 30.6  | 13.2 | 1.0  |       |      |
| (24) |                                                                   | 5% MCWB     | -7.5  | -3.2  | 16.8  | 23.9  | 32.1  | 49.8  | 55.4  | 54.3  | 46.0  | 29.0  | 12.5 | 0.4  |       |      |
| (25) |                                                                   | 10% DB      | -10.2 | -6.4  | 12.1  | 21.6  | 31.9  | 49.1  | 56.2  | 55.7  | 46.1  | 28.9  | 9.2  | -2.6 |       |      |
| (26) |                                                                   | 10% MCWB    | -10.6 | -6.9  | 11.1  | 20.2  | 30.8  | 45.9  | 52.7  | 52.2  | 43.8  | 27.5  | 8.4  | -3.3 |       |      |
| (27) |                                                                   | 0.4% WB     | 0.8   | 8.5   | 25.5  | 30.6  | 37.0  | 58.5  | 62.4  | 59.6  | 50.5  | 33.3  | 21.7 | 7.3  |       |      |
| (28) |                                                                   | 0.4% MCDB   | 1.4   | 9.1   | 26.7  | 32.9  | 40.3  | 65.4  | 66.7  | 65.2  | 53.4  | 34.4  | 22.0 | 7.9  |       |      |
| (29) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | -3.8  | 2.7   | 21.4  | 26.9  | 33.9  | 53.2  | 58.8  | 56.5  | 48.1  | 31.3  | 16.3 | 3.7  |       | (34) |
| (30) |                                                                   | 2% MCDB     | -3.4  | 3.4   | 22.4  | 28.7  | 35.6  | 58.3  | 63.7  | 60.6  | 50.0  | 32.5  | 17.0 | 4.2  |       |      |
| (31) |                                                                   | 5% WB       | -7.6  | -3.1  | 17.1  | 23.9  | 32.3  | 49.9  | 56.0  | 54.6  | 46.1  | 29.3  | 12.4 | 0.4  |       |      |
| (32) |                                                                   | 5% MCDB     | -7.3  | -2.6  | 18.2  | 25.5  | 33.6  | 53.6  | 59.7  | 58.1  | 48.4  | 30.5  | 13.0 | 1.0  |       |      |
| (33) |                                                                   | 10% WB      | -10.6 | -7.0  | 11.1  | 20.2  | 30.8  | 45.8  | 53.1  | 52.6  | 43.8  | 27.6  | 8.6  | -3.2 |       |      |
| (34) |                                                                   | 10% MCDB    | -10.3 | -6.5  | 11.9  | 21.5  | 31.9  | 48.7  | 56.5  | 55.8  | 46.2  | 28.8  | 9.2  | -2.7 |       |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 7.6   | 8.6   | 10.3  | 12.7  | 9.1   | 8.4   | 7.7   | 7.2   | 5.1   | 5.4   | 8.2  | 7.9  |       | (39) |
| (36) |                                                                   | 5% DB MCDBR | 10.9  | 12.9  | 11.7  | 16.5  | 11.9  | 16.3  | 13.8  | 13.0  | 9.0   | 4.7   | 10.2 | 11.2 |       |      |
| (37) |                                                                   | 5% DB MCWBR | 10.4  | 12.7  | 11.8  | 15.3  | 10.7  | 11.8  | 10.0  | 9.8   | 7.2   | 4.4   | 10.1 | 11.0 |       |      |
| (38) |                                                                   | 5% WB MCDBR | 10.5  | 12.8  | 12.2  | 16.2  | 10.9  | 15.8  | 13.5  | 12.2  | 8.7   | 4.4   | 10.4 | 11.2 |       |      |
| (39) |                                                                   | 5% WB MCWBR | 10.4  | 12.6  | 11.8  | 15.1  | 9.8   | 11.9  | 10.2  | 10.0  | 7.3   | 4.3   | 10.3 | 11.1 |       |      |
| (40) |                                                                   |             |       |       |       |       |       |       |       |       |       |       |      |      |       |      |
| (41) | Clear-Sky Solar Irradiance                                        | taub        | N/A   | 0.194 | 0.277 | 0.306 | 0.336 | 0.339 | 0.395 | 0.375 | 0.300 | 0.206 | N/A  | N/A  |       | (43) |
| (42) |                                                                   | taud        | N/A   | 1.944 | 2.017 | 2.052 | 2.090 | 2.311 | 2.278 | 2.339 | 2.448 | 2.015 | N/A  | N/A  |       |      |
| (43) |                                                                   | Ebn at Noon | 0     | 139   | 211   | 252   | 260   | 266   | 242   | 231   | 219   | 126   | 0    | 0    |       |      |
| (44) |                                                                   | Edh at Noon | 0     | 10    | 25    | 35    | 39    | 33    | 33    | 27    | 17    | 9     | 0    | 0    |       |      |
| (45) | All-Sky Solar Radiation                                           | RadAvg      | 0     | 69    | 533   | 1309  | 1874  | 1915  | 1453  | 890   | 431   | 119   | 0    | 0    |       | (45) |
| (46) |                                                                   | RadStd      | 0     | 8     | 25    | 50    | 79    | 132   | 124   | 103   | 44    | 17    | 0    | 0    |       |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (4 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|      |                        | +1.89 | N/A     | N/A    | N/A     | N/A   | N/A   | -743        | -709  | N/A   | N/A   |

Nomenclature: See separate page